

# Tutorato di algoritmi e laboratorio 7

## Longest Common Substring

alfabeto:

$$\Sigma = \{a, b, c, \dots, y, z\}$$

$$\Sigma^* = \{\epsilon, a, b, c, \dots, y, z, aa, ab, \dots, zz, aaa, \dots\}$$

$\epsilon$  := stringa vuota

stefano = estefano

longest common substring

$$\underline{\alpha} = \langle \overset{T}{\alpha_1}, \overset{U}{\alpha_2}, \dots, \overset{O}{\alpha_k} \rangle, \underline{\beta} = \langle \overset{L}{\beta_1}, \overset{E}{\beta_2}, \dots, \overset{E}{\beta_n} \rangle$$

$$\underline{LCS(\alpha, \beta)} = \langle \alpha_{i_1}, \alpha_{i_2}, \dots, \alpha_{i_m} \rangle = \langle \beta_{j_1}, \beta_{j_2}, \dots, \beta_{j_n} \rangle$$

dove  $i_x < i_y \Leftrightarrow x < y$  e  $m$  massimo!

FARE

E

A

AR

DARE

R

AE

E

AE

ARE

DIRE

E

EAN

una soluzione ricorsiva

base :  $LCS(\underline{\epsilon}, \underline{q}) = LCS(\underline{q}, \underline{\epsilon}) = \underline{0}$

passo ricorsivo :

$LCS(\langle \alpha_1, \dots, \alpha_n, \bar{\delta} \rangle, \langle \beta_1, \dots, \beta_n, \bar{\delta} \rangle)$

=

$LCS(\langle \alpha_1, \dots, \alpha_n \rangle, \langle \beta_1, \dots, \beta_n \rangle) + 1$

se  $\underline{\delta} = \underline{\delta}$

max

$LCS(\langle \alpha_1, \dots, \alpha_{n-1} \rangle, \langle \beta_1, \dots, \beta_n \rangle)$

$LCS(\langle \alpha_1, \dots, \alpha_n \rangle, \langle \beta_1, \dots, \beta_{n-1} \rangle)$

se  $\delta \neq \delta$

CIAO  
E

E

1000000

2AIND  
VASO

$\delta = \delta$

$\delta \neq \delta$

2AIND  
COSA

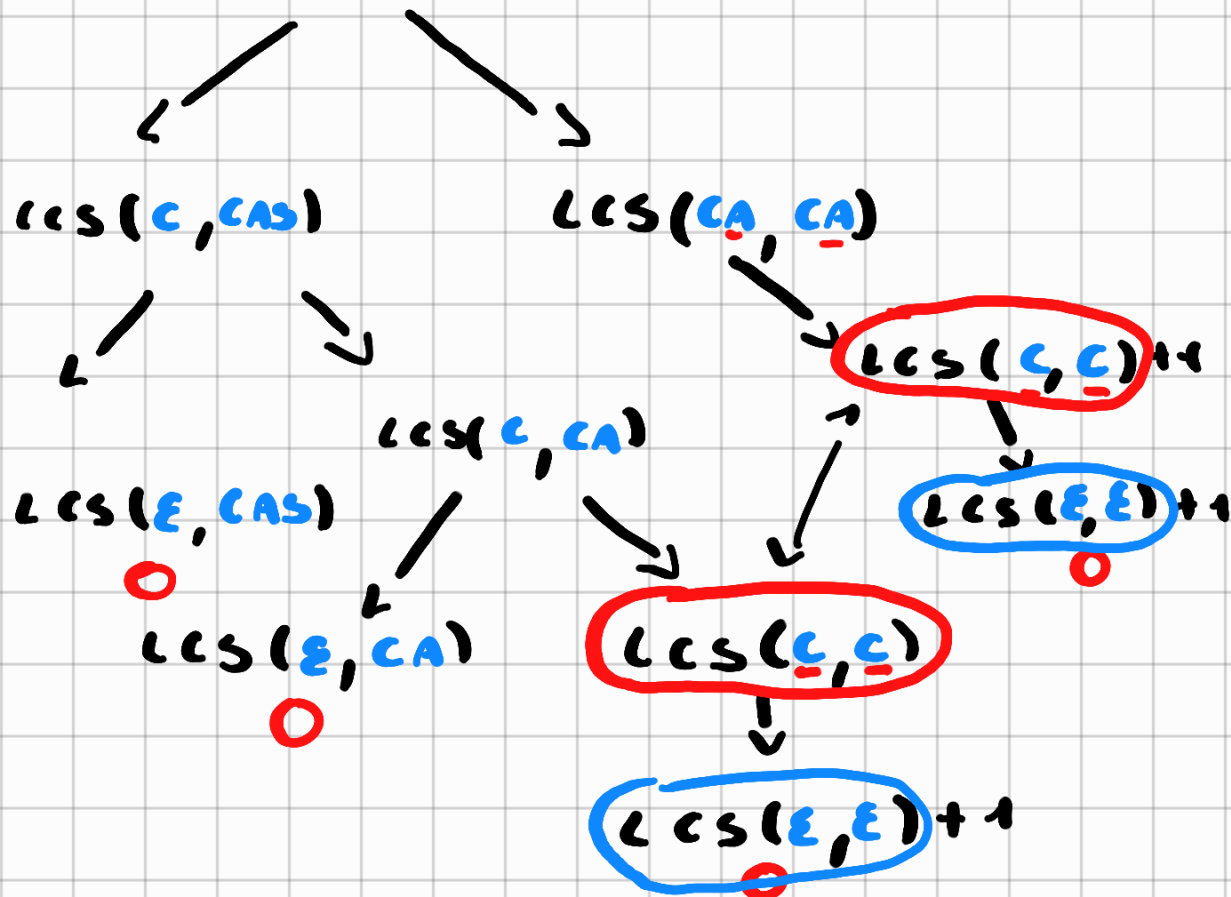
Svolgiamo un esempio:

$LCS(CANE, CASE)$

$LCS(CAN, CAS) + 1$

$LCS(CA, CAS)$

$LCS(CAN, CA)$



# Programmazione dinamica

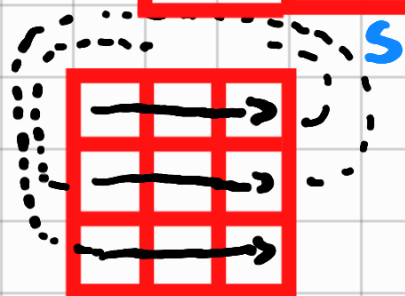
LCS(STEFANO, SAVINO)

STEFANO = ESTEFANO

SAVINO = SAVINO

0 1 2 3 4 5 6 7  
E S T E F A \_ N O

|     |   |   |   |   |   |   |   |   |
|-----|---|---|---|---|---|---|---|---|
| 0 E | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 S | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 A | 0 | 1 | 1 | 1 | 1 | 2 | 2 | 2 |
| 3 V | 0 | 1 | 1 | 1 | 1 | 2 | 2 | 2 |
| 4 I | 0 | 1 | 1 | 1 | 1 | 2 | 2 | 2 |
| 5 _ | 0 | 1 | 1 | 1 | 1 | 2 | 3 | 3 |
| 6 O | 0 | 1 | 1 | 1 | 1 | 2 | 3 | 4 |



$$A_{ij} = \text{len}(\text{LCS}(\langle \alpha_1, \dots, \alpha_i \rangle, \langle \beta_1, \dots, \beta_j \rangle))$$

LCS = SAVINO

LCS(SAIT, SIAT)

|   | E | S | A | I | T |
|---|---|---|---|---|---|
| E | 0 | 0 | 0 | 0 | 0 |
| S | 0 | 1 | 1 | 1 | 1 |
| I | 0 | 1 | 1 | 2 | 2 |
| A | 0 | 1 | 2 | 2 | 2 |
| T | 0 | 1 | 2 | 2 | 3 |

S A T

S I T

↖ ↘

# Bellman Ford

$$G = (V, E)$$

$$V = \{a, b, c, d\}$$

$$E = \{ \underbrace{(d, c, 2), (c, b, 1), (b, d, 8), (c, d, 1)}_{\rightarrow}, (b, c, 1), (a, b, 2) \}$$

$T_0 \quad T_1 \quad T_2 \quad T_3$

$d[a]$

|   |   |   |   |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
|---|---|---|---|

$d[b]$

|                        |   |   |   |
|------------------------|---|---|---|
| $\infty \rightarrow 2$ | 2 | 2 | 2 |
|------------------------|---|---|---|

$d[c]$

|          |                        |   |   |
|----------|------------------------|---|---|
| $\infty$ | $\infty \rightarrow 3$ | 3 | 3 |
|----------|------------------------|---|---|

$d[d]$

|          |                         |                    |   |
|----------|-------------------------|--------------------|---|
| $\infty$ | $\infty \rightarrow 10$ | $10 \rightarrow 4$ | 4 |
|----------|-------------------------|--------------------|---|

$\infty + 2 \quad \infty \quad X$

$\infty + 1 \quad \infty \quad X$

$\infty + 8 \quad \infty \quad X$

$\infty + 1 \quad \infty \quad X$

$\infty + 1 \quad \infty \quad X$

$\rightarrow 0 + 2 \quad \infty \quad \checkmark$

$\infty + 2 \quad \infty \quad X$

$\infty + 1 \quad 2 \quad X$

$2 + 8 \quad \infty \quad \checkmark$

$\infty + 1 \quad 10 \quad X$

$2 + 1 \quad \infty \quad \checkmark$

$\rightarrow 0 + 2 \quad 2 \quad X$

$10 + 2 \quad 3 \quad X$

$3 + 1 \quad 2 \quad X$

$2 + 8 \quad 10 \quad X$

$3 + 1 \quad 10 \quad \checkmark$

$2 + 1 \quad 3 \quad X$

$\rightarrow 0 + 2 \quad 2 \quad X$

# Floyd - Warshall

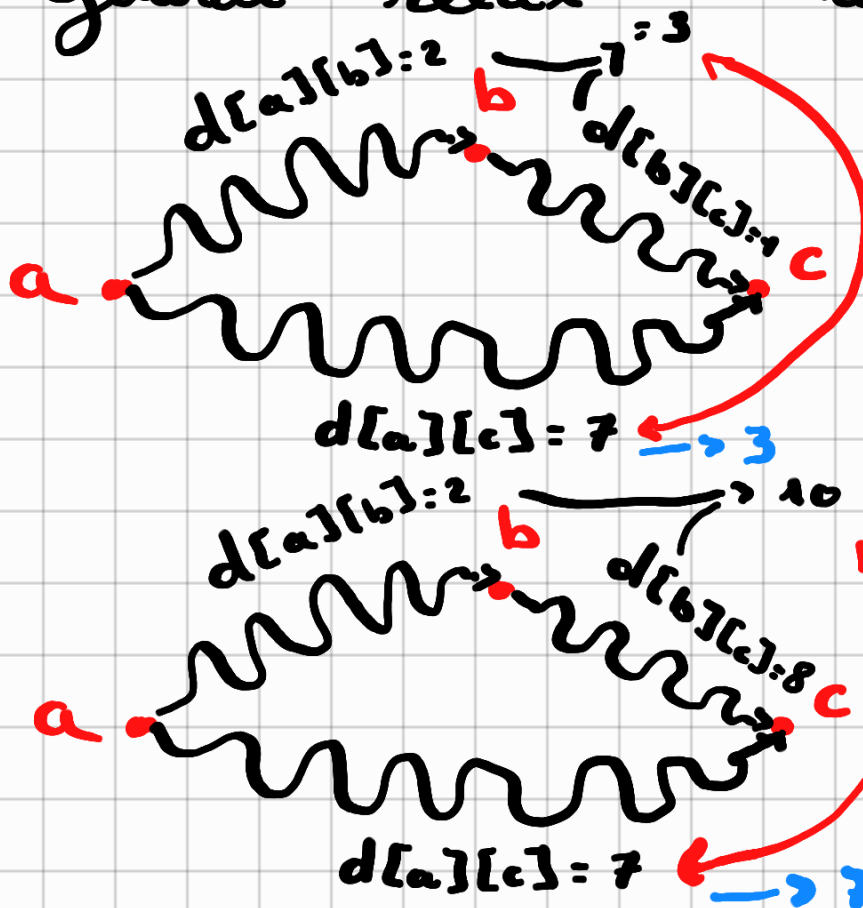
all pair :

al tempo 0 :

$$d[u][v] = \begin{cases} 0 & \text{se } u = v \\ w(u,v) & \text{se esiste } (u,v) \in E \\ \infty & \text{altrimenti} \end{cases}$$

$$p[u][v] = \begin{cases} \text{NIL} & \text{se } u = v \\ u & \text{se esiste } (u,v) \in E \\ \text{NIL} & \text{altrimenti} \end{cases}$$

General relax



relax:

$$\text{if } d[v] > d[u] + w(u,v) \\ d[v] = d[u] + w(u,v)$$

g-relax:

$$\text{if } d[u,v] > d[u,k] + d[k,v] \\ d[u,v] = d[u,k] + d[k,v]$$

Floyd-Warshall ( $G, w$ )

for  $b \in V$

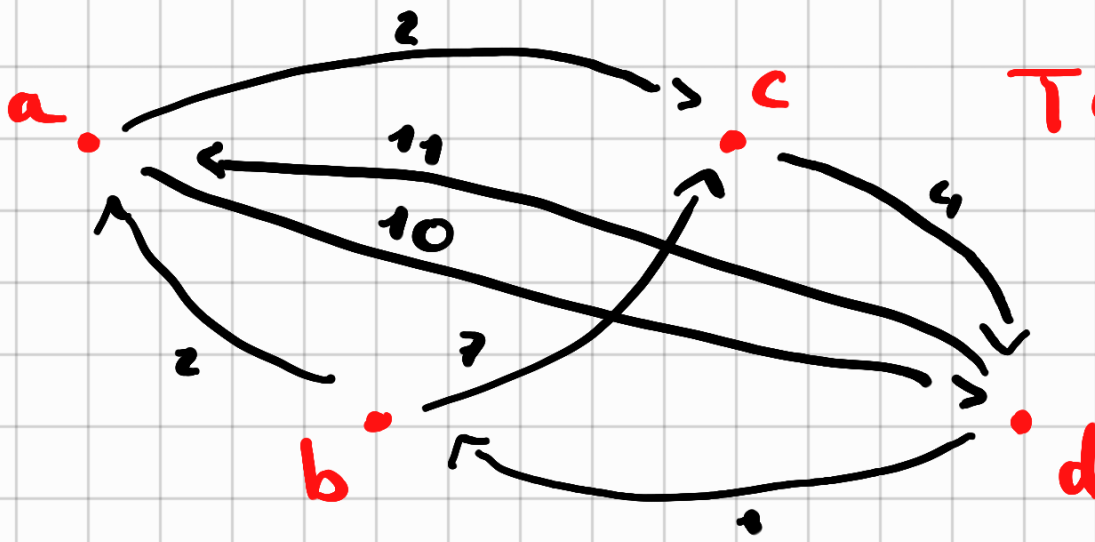
for  $a \in V$

for  $c \in V$

g-relax( $a, c, b$ )

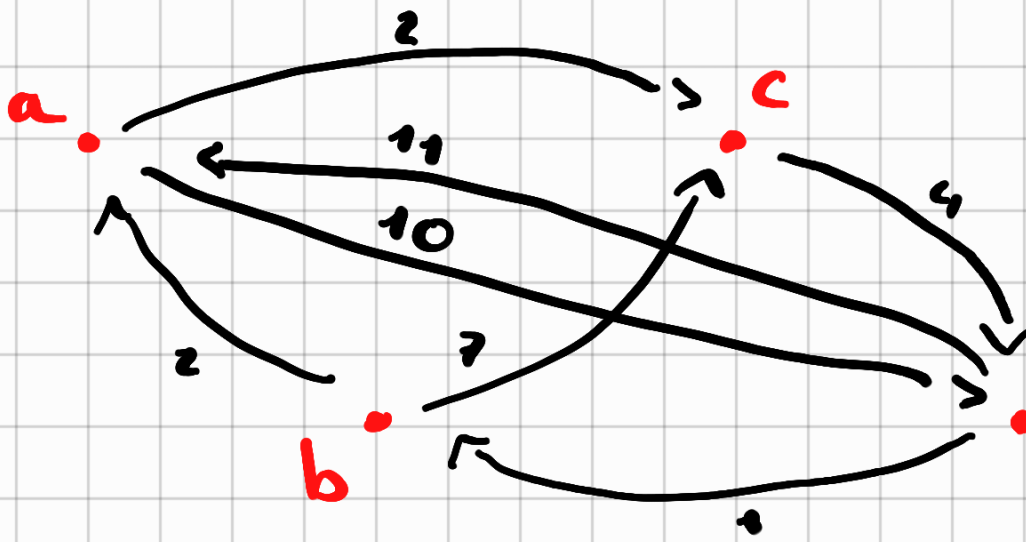


TEMPO 0



|   | d        | a        | b        | c        | d |
|---|----------|----------|----------|----------|---|
| a | 0        | $\infty$ | 2        | 10       |   |
| b | 2        | 0        | 7        | $\infty$ |   |
| c | $\infty$ | $\infty$ | 0        | 4        |   |
| d | 11       | 1        | $\infty$ | 0        |   |

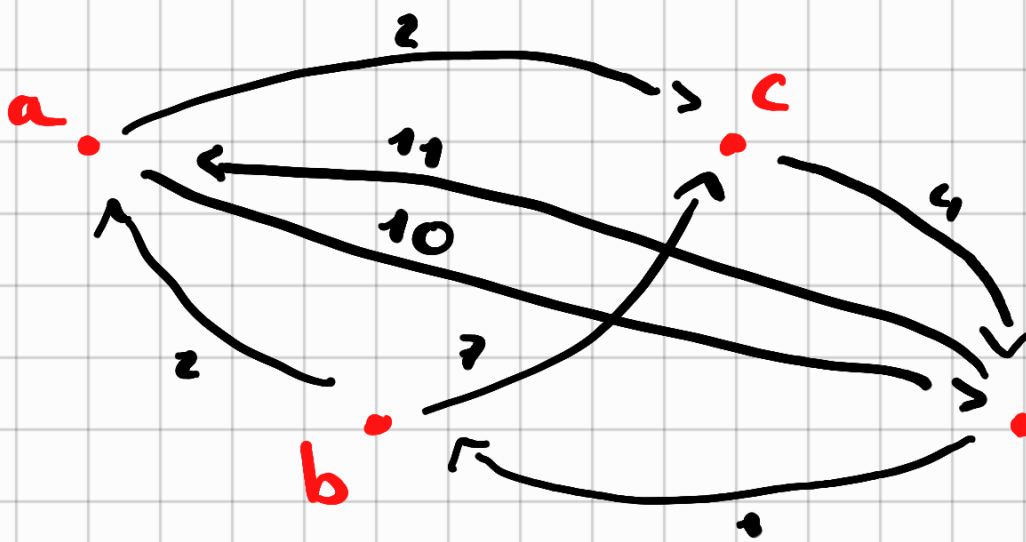
|   | d   | a   | b   | c   | d |
|---|-----|-----|-----|-----|---|
| a | nil | nil | a   | a   |   |
| b | b   | nil | b   | nil |   |
| c | nil | nil | nil | c   |   |
| d | d   | d   | nil | nil |   |



TEMPO 1  
 rilassiamo  
 su a  
 proviamo  
 a passare  
 d sempre da a

| d | a        | b        | c  | d  | P |
|---|----------|----------|----|----|---|
| a | 0        | $\infty$ | 2  | 10 | a |
| b | 2        | 0        | 4  | 12 | b |
| c | $\infty$ | $\infty$ | 0  | 4  | c |
| d | 11       | 1        | 13 | 0  | d |

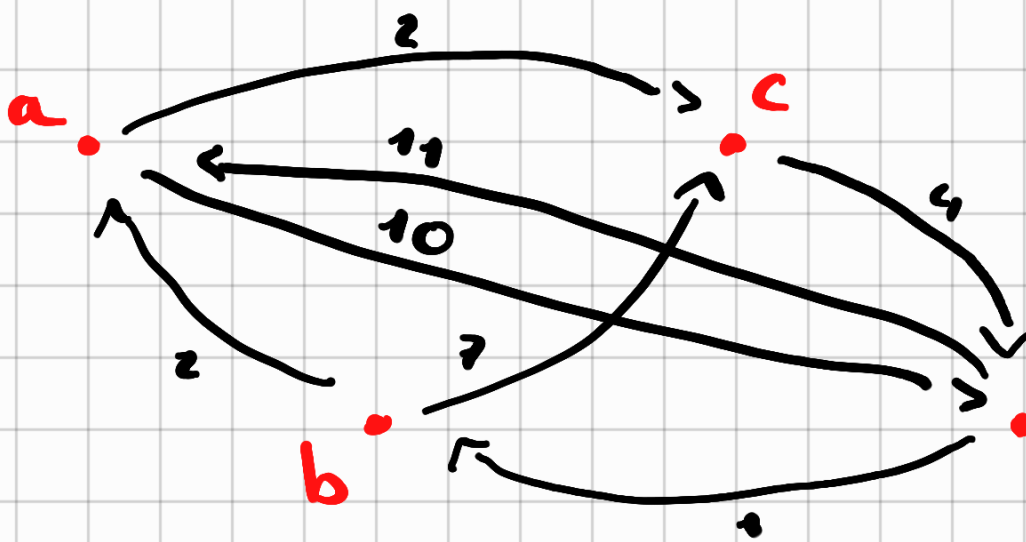
|   | a   | b   | c   | d   |
|---|-----|-----|-----|-----|
| a | Nil | Nil | a   | a   |
| b | b   | Nil | a   | a   |
| c | Nil | Nil | Nil | c   |
| d | d   | d   | a   | Nil |



TEMPO 2  
 rilassiamo  
 su b  
 proviamo  
 a passare  
 d sempre da a

| d | a        | b        | c | d  | P |
|---|----------|----------|---|----|---|
| a | 0        | $\infty$ | 2 | 10 | a |
| b | 2        | 0        | 4 | 12 | b |
| c | $\infty$ | $\infty$ | 0 | 4  | c |
| d | 3        | 1        | 5 | 0  | d |

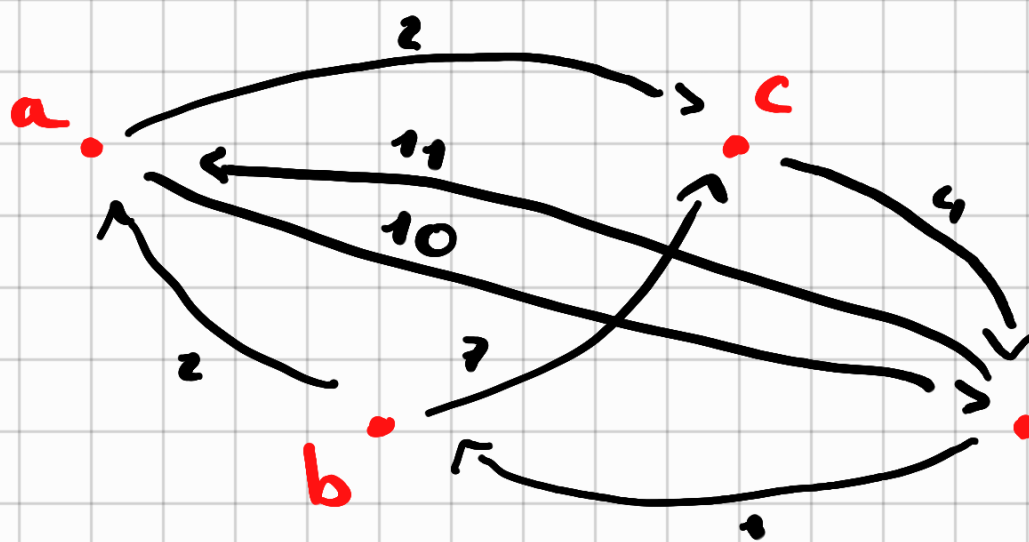
|   | a   | b   | c   | d   |
|---|-----|-----|-----|-----|
| a | Nil | Nil | a   | a   |
| b | b   | Nil | a   | a   |
| c | Nil | Nil | Nil | c   |
| d | b   | d   | b   | Nil |



TEMPO 3  
rilasciamo  
su c  
proviamo  
a passare  
d sempre da a

|   | d        | a        | b | c | d |
|---|----------|----------|---|---|---|
| a | 0        | $\infty$ | 2 | 6 |   |
| b | 2        | 0        | 4 | 8 |   |
| c | $\infty$ | $\infty$ | 0 | 4 |   |
| d | 3        | 1        | 5 | 0 |   |

|   | d   | a   | b   | c   | d |
|---|-----|-----|-----|-----|---|
| a | Nil | Nil | a   | c   |   |
| b | b   | Nil | a   | c   |   |
| c | Nil | Nil | Nil | c   |   |
| d | b   | d   | b   | Nil |   |



TEMPO 4  
rilasciamo  
su d  
proviamo  
a passare  
d sempre da a

|   | d | a | b | c | d | p | a   | b   | c   | d   |
|---|---|---|---|---|---|---|-----|-----|-----|-----|
| a | 0 | 7 | 2 | 6 |   | a | nil | d   | a   | c   |
| b | 2 | 0 | 4 | 8 |   | b | b   | nil | a   | c   |
| c | 7 | 5 | 0 | 4 |   | c | d   | d   | nil | c   |
| d | 3 | 1 | 5 | 0 |   | d | b   | d   | b   | nil |

Cammino minimo da a a d

$\langle a, c, d \rangle$  6

Cammino minimo da a a b

$\langle a, c, d, b \rangle$  7